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Studierichting: Informatica
Oefeningenreeks 4A nr. 39.4

$$\int \frac{3x+1}{(x^2-4x+5)^2}dx$$
$$= \int \frac{3x+1}{((x-2)^2+1)^2}$$
$$t = x - 2 \Rightarrow dt = dx$$
$$\Rightarrow \int \frac{3t+7}{(t^2+1)^2}dt$$
$$= 3\int \frac{t}{(t^2+1)^2}dt + 7\int \frac{dt}{(t^2+1)^2}$$
$$A = 3\int \frac{t}{(t^2+1)^2}dt = \frac{3}{2}\int (t^2+1)^{-2}d(t^2+1) = -\frac{3}{2(t^2+1)}$$
$$B = 7\int \frac{1}{(t^2+1)^2}dt = 7\left(\frac{t}{2(t^2+1)} + \frac{1}{2}\int \frac{1}{t^2+1}dt\right)$$
$$= \frac{7t}{2(t^2+1)} + \frac{7}{2}\operatorname{Bgtan}(t)$$
$$A + B = \frac{7t-3}{2(t^2+1)} + \frac{7}{2}\operatorname{Bgtan}(t) + C$$
$$= \frac{7(x-2)-3}{2((x-2)^2+1)} + \frac{7}{2}\operatorname{Bgtan}(x-2) + C$$
$$= \frac{7x-17}{2(x^2-4x+5)} + \frac{7}{2}\operatorname{Bgtan}(x-2) + C$$

References